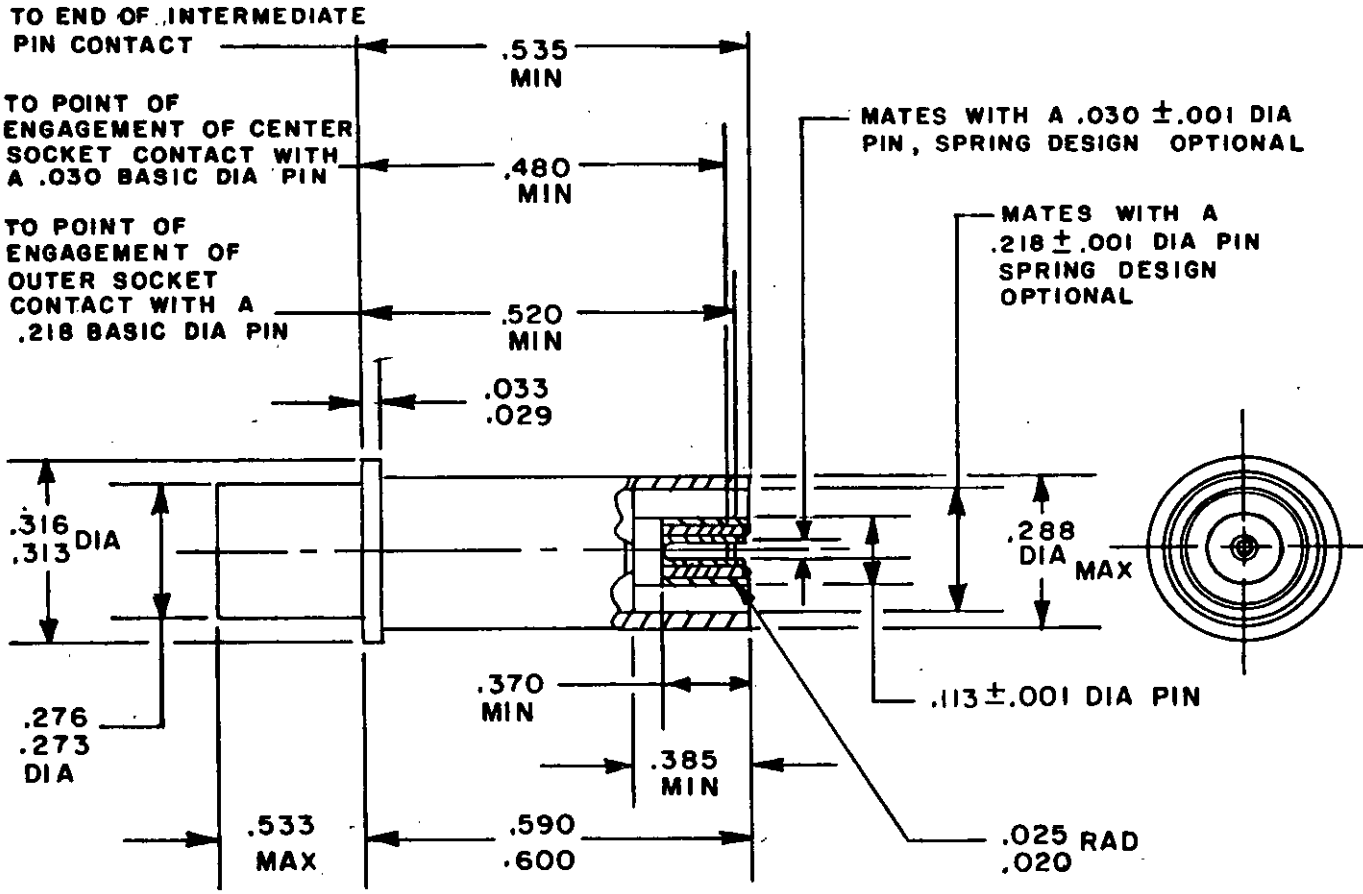
MILITARY SPECIFICATION SHEET

CONTACT, ELECTRICAL CONNECTOR,
CONCENTRIC TWINAX, SOCKET, SHIELDED, SIZE 8
(FOR DOD-C-83527, CONNECTOR)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the contact described herein shall consist of this specification and the latest issue of MIL-C-39029.



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	0.03	.113	2.87	.316	8.03	.533	13.54
.020	0.51	.218	5.54	.370	9.40	.535	13.59
.025	0.64	.273	6.93	.385	9.78	.590	14.99
.030	0.76	.276	7.01	.480	12.19	.600	15.24
.113	2.87	.313	7.95	.520	13.21		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.

FIGURE 1. Socket contact.

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REQUIREMENTS:

Qualification: Contacts shall comply with reliability assurance provisions of MIL-STD-790 as specified in MIL-C-38999.

Design and construction:

Dimensions and configuration: See figures 1 through 4 and table I.

TABLE I. Design characteristics.

BIN code	Color bands			Contact cavity size	Cable accommodated	Type	Class
	1st	2nd	3rd				
545	Green	Yellow	Green	8	M17/176-00002	E	B

Assembly procedure: Manufacturer's recommended assembly instructions shall be shipped with unit package.

Material:

Heat shrink boot: Shall be in accordance with MIL-I-23053/8.

Electrical:

Low signal level contact resistance (center contact only): See table II.

Contact resistance: See table III.

Frequency: 0 to 20 MHz (operating frequency range).

Voltage rating: 500 volts rms maximum; working voltage at sea level, 125 volts rms maximum at 70,000 feet.

TABLE II. Low signal level contact resistance (center contact only).

BIN code	Cable accommodated	Maximum contact resistance (milliohms)	
		Initial	After conditioning
545	M17/176-00002	55	66

TABLE III. Contact resistance.

BIN code	Contact	Cable accomodated	Test current (amperes)	Maximum voltage drop (millivolts)		
				25°C, +3°C, -0°C		+175°C, +3°C, -0°C
				Initial	After conditioning	After conditioning
545	Center	M17/176-00002	1.0	55	66	94
545	Intermediate	M17/176-00002	1.0	55	66	94
545	Outer	M17/176-00002	12.0	75	90	128

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Dielectric withstanding voltage: Shall be as specified in table IV.

TABLE IV. Dielectric withstanding voltage.

Contacts	Altitude	Test voltage ac rms
Center to intermediate	Sea level	1000
Intermediate to outer	Sea level	500

Mating contact: Shall be in accordance with MIL-C-39039/96.

Mechanical:

Contact engagement and separation force (socket contacts only): The engagement depth shall be a minimum of 0.7 of the minimum socket bored. The test pins shall be in accordance with MS3197 except the diameters shall be as specified in table V.

TABLE V. Contact engagement and separation force.

Test pin diameter (inch)	Minimum separation force (ounces)		Maximum engagement force (ounces		Maximum average engagement force
	Initial	After conditioning	Initial	After conditioning	
.2190 +.0000 -.0002	NA	NA	48	60	NA
.2170 +.0002 -.0000	4.0	2.0	NA	NA	NA
.0305 +.0000 -.0002	NA	NA	12	14	NA
.0295 +.0002 -.0000	0.5	0.4	NA	NA	NA

Crimp tensile strength (center, intermediate, and outer contact crimp joint): See table VI.

TABLE VI. Crimp tensile strength (at ambient).

BIN code	Cable accommodated	Axial load (pounds, minimum)
545	M17/176-00002	25

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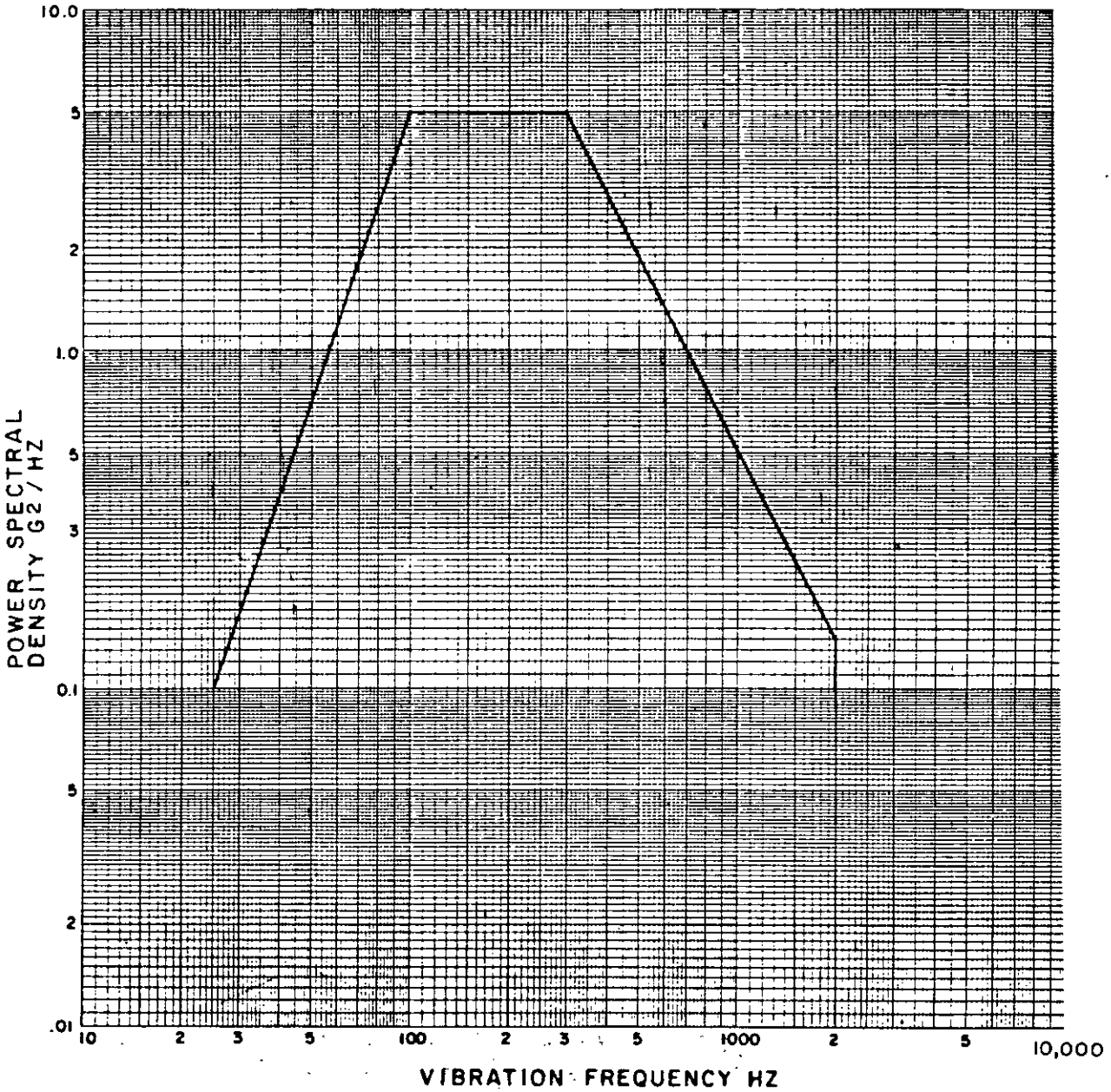


FIGURE 2. Vibration envelope.

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Vibration: Method 2005 of MIL-STD-1344, test condition V. The following details shall apply:

a. Use the vibration envelope shown on figure 5.

b. Vibration: To be conducted at standard test conditions.

c. Duration: 8 hours in the longitudinal direction and 8 hours in a perpendicular direction (16 hours).

High impact shock: All contacts shall be wired in a series circuit with 100 milliamperes maximum current flow through the series circuit during shock. Connectors shall be monitored for any discontinuities. A detector capable of detecting all discontinuities in excess of 1 microsecond shall be used. Wired and mated connectors shall be subjected to the test specified in MIL-S-901, grade A with the following modifications and additions. Mounting fixture shall be in accordance with MIL-S-901, lightweight. The cable or wire bundle shall be supported on a stationary frame to provide a free flexing cable length between the frame and fixture of not less than 36 inches (914.4 mm).

a. *Coupling: Coupled by normal mounting means.*

b. Test condition A: Plug shall be terminated with at least 80 percent of wired contacts. The wire bundle shall be provided with straight, open frame, strain relief accessory hardware.

Tools: Shall be as specified in tables VII and VIII.

TABLE VII. Tools (installing and removal).

BIN code	Installing and removal tool	Removal tool
545	M81969/14-06	M81969/14-06
545	M81969/46-06	M81969/46-12

TABLE VIII. Tools (crimping).

	Center contact tooling		Intermediate contact tooling		Outer crimp sleeve tooling	
BIN code	Basic crimping tool	Contact positioner	Basic crimping tool	Contact positioner	Basic crimping tool	Die part no.
545	M22520/2-01	M22520/2-37 (Proposed)	M22520/5-01 (Proposed)	M22520/5-200 (Proposed)	M22520/5-01 (Proposed)	M22520/5-200 (Proposed)

Environmental:

Thermal shock: High temperature shall be +85°C except for connectors using +200°C cables; high temperature shall then be +150°C.

Temperature rating: -65°C to +150°C.

Humidity: Perform humidity upon completion of group 1, using qualified MIL-C-38999H connectors with the 25-20 insert arrangement. When tested as specified below, wired, mated contacts shall show no deterioration which will adversely affect the performance of the contact. Following the test and during the final cycle, insulation resistance shall exceed 100 megohms at +25°C. Insulation resistance shall be measured between the crimped outer and intermediate contacts and between the crimped inner and intermediate contacts in accordance with method 3003 of MIL-STD-1344.

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Humidity test: Wired, mated connectors shall be subjected to the humidity test specified in method 1002 of MIL-STD-1344. The following details and exceptions shall apply:

a. Test condition: Type II omitting subcycle step 7 (b).

b. The mated connectors shall be mounted in a vertical position.

c. Step 7 (a) shall be performed during the last cycle.

d. Three hours minimum after the start of step 7 (a) during the final cycle and while the connectors are still subjected to high humidity, the insulation resistance shall be measured when the chamber temperature reaches 20°C ±5°C and condensation is observed on the connector.

Part number: M39029/95-545.

QPL evaluating activity: Defense Electronics Supply Center (DESC-E), Dayton, OH 45444.

Custodians:
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Navy - AS
Air Force - 85

Preparing activity:
Air Force - 85

(Project 5935-3415-3)

Review activities:
Navy - EC, SH, MC
Air Force - 11, 15, 17, 99
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User activities:
Army - AT
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